

Objectives

- To manage the project workflow and ensure the data pipelines are optimized for the subsequent Machine Learning phases by the team.
- To transform a large-scale historical dataset of Philippine food prices (2.5 million rows) into boolean transaction baskets suitable for Association Rule Mining (ARM).
- To engineer two distinct transaction types: Market-Day Co-Occurrence baskets and Regional Price-Spike baskets, to identify both physical supply chain patterns and inflationary contagion.

Tasks Completed

Leader & Data Architect (Jayson Buenosaires)

- **Tasks:** Project management, data loading, and transaction engineering.
- **Description:**
 - As the team leader, I managed the ingestion and structural analysis of the master dataset containing over 2.49 million records, isolating the 2018–2021 window to ensure transaction sizes remained mathematically manageable for the ML engineers.
 - Engineered 46,591 Co-Occurrence baskets using Pandas `groupby` functions to track commodities sold in the exact same market on the exact same day.
 - Developed the statistical logic for Price Spike baskets. By calculating a per-commodity 75th percentile baseline to flag anomalous price shocks, I successfully generated 38,053 highly relevant spike transactions ready for ARM analysis.
- **Screenshot/Code Snippet:**

Python

```
# Calculating per-commodity 75th percentile to normalize across scales
```

```
p75 = (  
    df.groupby('item')['mp_price']  
    .quantile(0.75)  
    .rename('p75')  
    .reset_index()  
)
```

```
df_sp = df.merge(p75, on='item')  
df_sp['is_spike'] = df_sp['mp_price'] >= df_sp['p75']
```

```
# Engineering the Price Spike baskets by region and date
```

```
spike_baskets = (  
    df_sp[df_sp['is_spike']]  
    .groupby(['adm1_name', 'date'])['item']  
    .apply(lambda x: list(set(x)))  
    .reset_index(name='items')  
)
```

Tasks to be Completed

- Oversee the translation of the extracted high-lift price spike rules into a standardized "Warning Matrix" for local government units (LGUs).
- Coordinate with the team to finalize the Chapter 9 research manuscript, ensuring all data visualizations and algorithm comparisons align with the initial project objectives.

Results and Analysis

1. **Data Engineering Success:** The primary result of my phase was the successful transformation of raw time-series data into clean, aggregated arrays. By strictly defining what constitutes a "price spike" (using the 75th percentile rather than raw price), I ensured the downstream algorithms were looking for actual inflationary contagion rather than standard seasonal price variances.
2. **Foundation for Algorithmic Scaling:** The data architecture I established allowed the ML team to test the boundaries of different algorithms. The 46,591 Co-Occurrence baskets and 38,053 Price Spike baskets I engineered served as the direct foundation that allowed the FP-Growth model to successfully extract over 4.5 million rules.
3. **Project Management:** By splitting the transaction engineering (my role) from the algorithm optimization, the team was able to diagnose and bypass the severe `MemoryError` bottleneck caused by the Apriori algorithm much faster, successfully pivoting to FP-Growth.

Challenges and Solutions

- **Challenge:** The sheer volume of the dataset (2.49 million rows) threatened to overwhelm standard memory limits before the data even reached the algorithm stage, making the arrays too large for the `TransactionEncoder`.
- **Solution:** I implemented a strategic temporal filter (using only the 2018–2021 window) and utilized highly efficient `groupby` and `apply(set)` operations in Pandas to compress millions of raw rows into roughly ~84,000 distinct, manageable transaction baskets. This ensured a stable, memory-safe handoff to the ML engineers.

Conclusion

The data architecture phase of the pipeline was a complete success. I was able to successfully structure a massive agricultural dataset into a format that allows Machine Learning algorithms to map out complex economic patterns. As project leader, ensuring the integrity of these initial Co-Occurrence and Price Spike baskets was critical, as it directly enabled the rest of the team to identify actionable intelligence regarding food security and regional inflation across the Philippines.

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